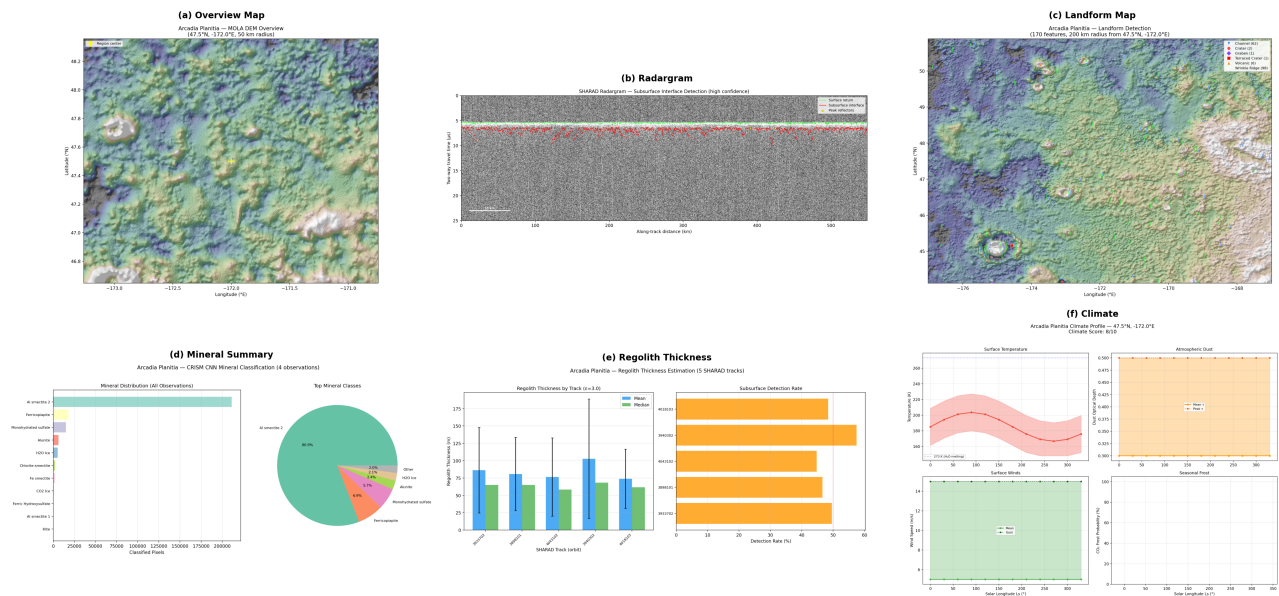


Arcadia Planitia — Comprehensive Multi-Instrument Scientific Report

Generated: 2026-02-19 11:49:04 UTC **Center:** 47.5°N, -172.0°E **Scan radius:** 200 km (landform), 50 km (detail) **Instruments:** MOLA DEM, SHARAD, CRISM, Mars Climate Model

Arcadia Planitia — Comprehensive Multi-Instrument Analysis

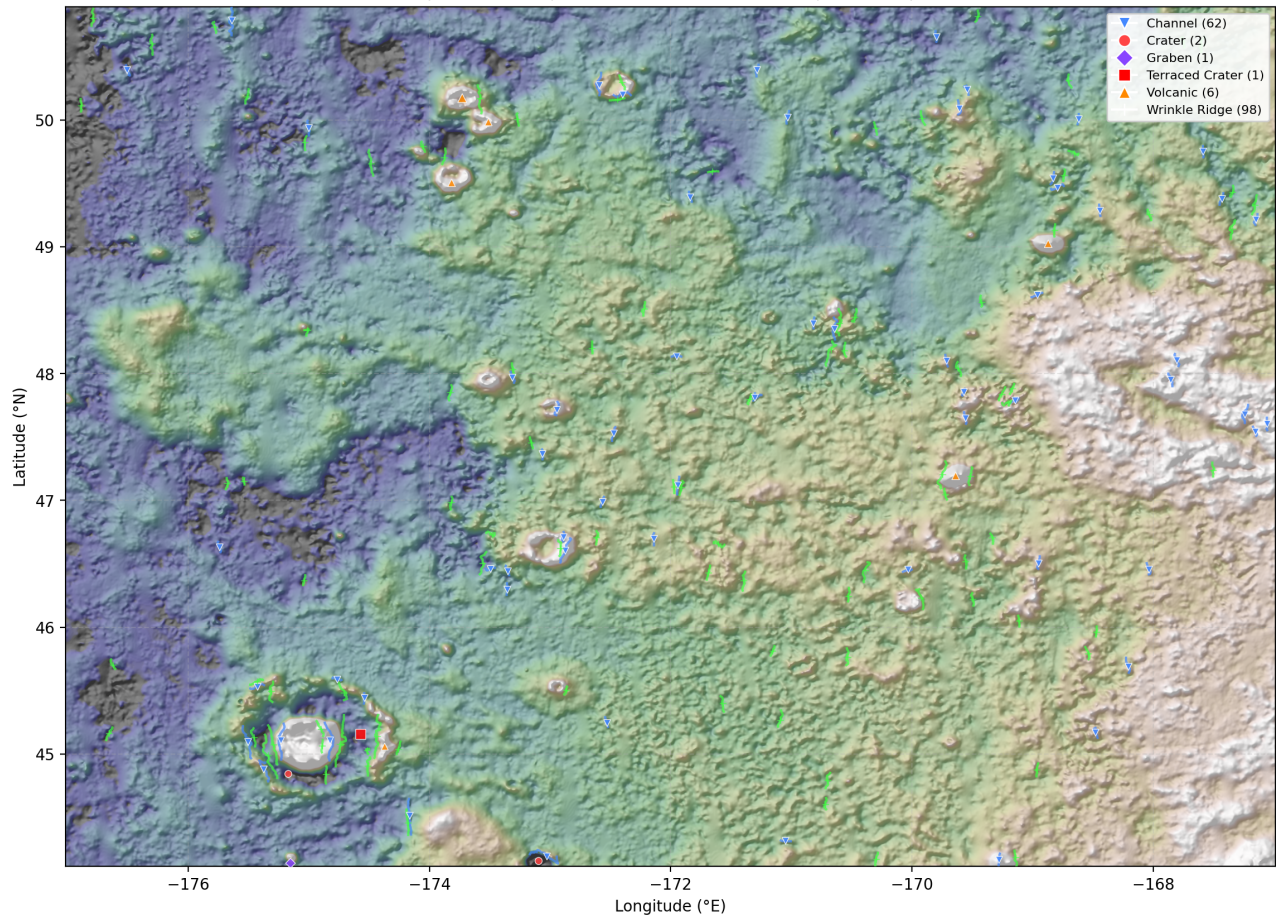


1. Landform Detection (MOLA DEM)

Total features detected: 170

Feature Type	Count
Craters	3
Volcanics	6
Graben	1
Channels	62
Ridges	98
Ldas	0

Arcadia Planitia — Landform Detection
(170 features, 200 km radius from 47.5°N, -172.0°E)

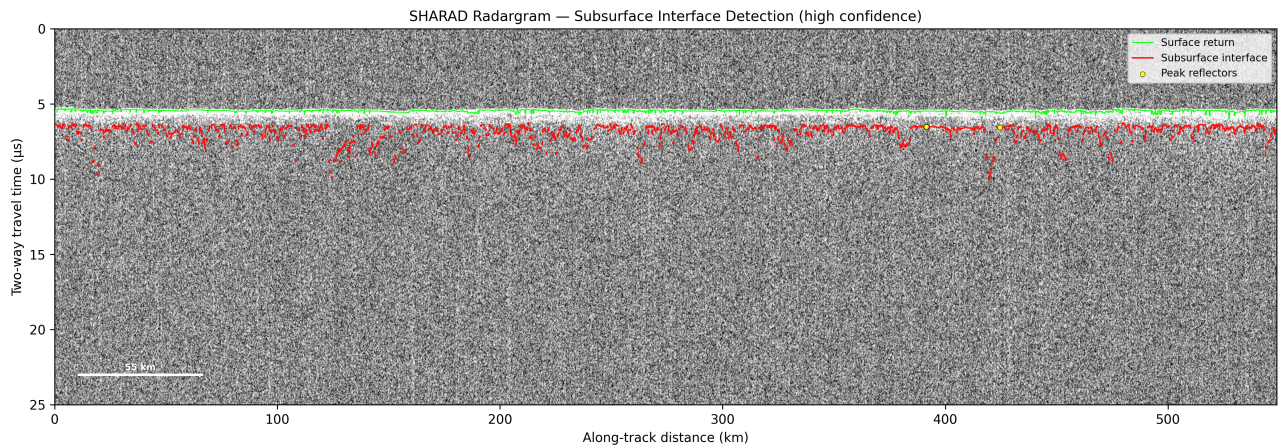


Notable Features

Type	Lat	Lon	Size	Depth/Height	Confidence	Description
terraced_crater	45.16	-174.57	15.3 km	96 m	0.242	Terraced crater (1 bench, terrace depth 20 m), d/D=0.0063, c
volcanic	50.18	-173.73	10.9 km	109 m	0.244	Volcanic construct (10.9 km), mean flank slope 2.0 deg, reli
volcanic	49.03	-168.87	8.2 km	186 m	0.591	Volcanic construct (8.2 km), mean flank slope 3.1 deg, relie
volcanic	47.20	-169.64	8.1 km	168 m	0.592	Volcanic construct (8.1 km), mean flank slope 3.2 deg, relie
volcanic	49.50	-173.82	7.8 km	132 m	0.201	Volcanic construct (7.8 km), mean flank slope 2.0 deg, relie
volcanic	49.99	-173.51	7.5 km	62 m	0.320	Volcanic construct (7.5 km), mean flank slope 1.3 deg, relie
crater	44.16	-173.10	7.0 km	102 m	0.514	Simple crater, d/D=0.0146, circularity=0.892
volcanic	45.06	-174.37	6.3 km	77 m	0.341	Volcanic construct (6.3 km), mean flank slope 2.0 deg, relie
crater	44.84	-175.17	5.2 km	41 m	0.448	Simple crater, d/D=0.0078, circularity=0.859
graben	44.14	-175.15	-	136 m	0.741	Graben, 20.6 km x 3.3 km, aspect ratio 6.2, depth 136 m
channel	44.51	-174.17	-	0 m	0.529	Channel/valley: length 17.1 km, width ~1.1 km, depth ~0 m, s
channel	44.19	-173.03	-	0 m	0.409	Channel/valley: length 17.7 km, width ~4.1 km, depth ~0 m, s
channel	45.10	-175.51	-	0 m	0.394	Channel/valley: length 17.1 km, width ~3.1 km, depth ~0 m, s
channel	44.88	-175.37	-	0 m	0.394	Channel/valley: length 14.0 km, width ~3.1 km, depth ~0 m, s
channel	50.02	-171.03	-	0 m	0.377	Channel/valley: length 6.0 km, width ~1.0 km, depth ~0 m, si
channel	50.79	-175.64	-	0 m	0.372	Channel/valley: length 14.4 km, width ~2.8 km, depth ~0 m, s
channel	49.29	-168.44	-	0 m	0.365	Channel/valley: length 5.2 km, width ~0.9 km, depth ~0 m, si
channel	47.79	-169.15	-	0 m	0.361	Channel/valley: length 6.8 km, width ~1.5 km, depth ~0 m, si
channel	47.12	-171.94	-	0 m	0.353	Channel/valley: length 8.3 km, width ~1.5 km, depth ~0 m, si
channel	49.38	-167.43	-	0 m	0.352	Channel/valley: length 5.6 km, width ~1.1 km, depth ~0 m, si

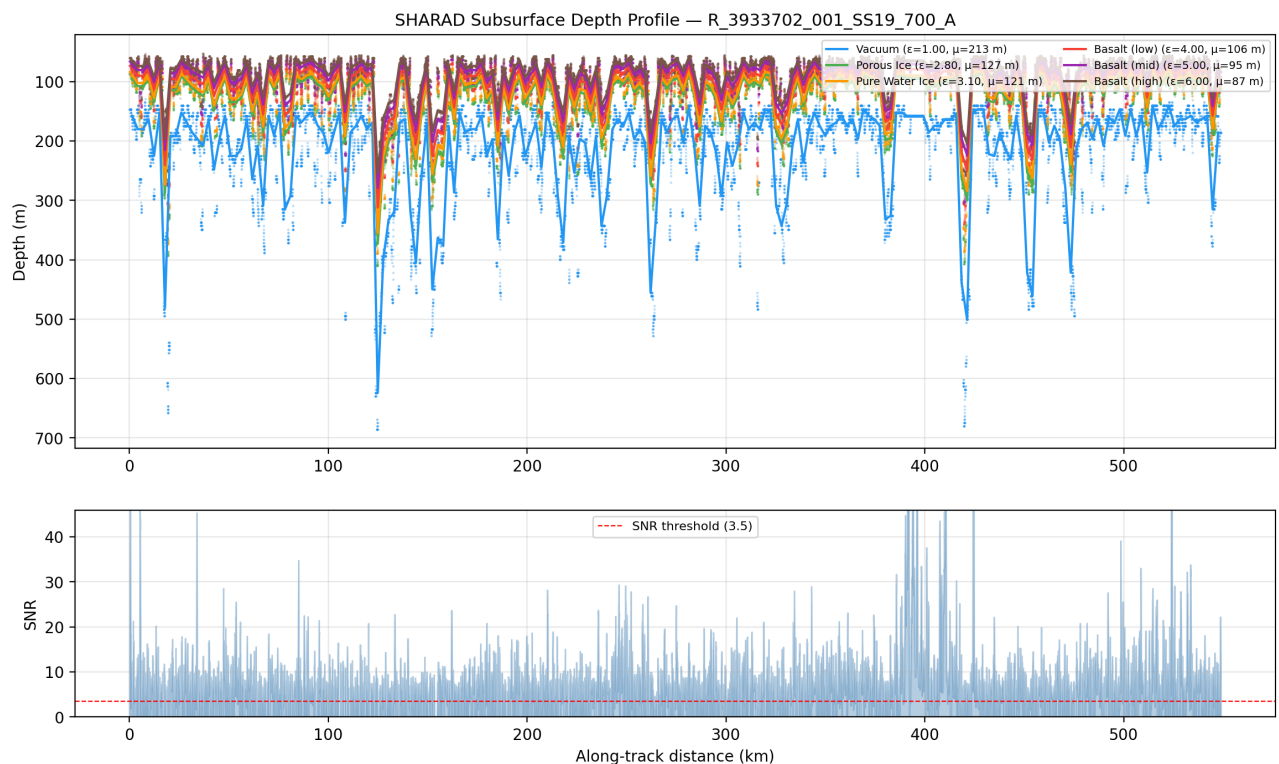
2. SHARAD Subsurface Interface Detection

Track: R_3933702_001_SS19_700_A **Traces:** 30,617 **Detection rate:** 81.3% **Confidence:** high



Depth Estimates by Dielectric Model

Material	ϵ_r	Mean Depth (m)	Min (m)	Max (m)
Vacuum	1.00	212.7	129.3	685.8
Porous Ice	2.80	127.1	77.3	409.8
Pure Water Ice	3.10	120.8	73.4	389.5
Basalt (low)	4.00	106.3	64.6	342.9
Basalt (mid)	5.00	95.1	57.8	306.7
Basalt (high)	6.00	86.8	52.8	280.0

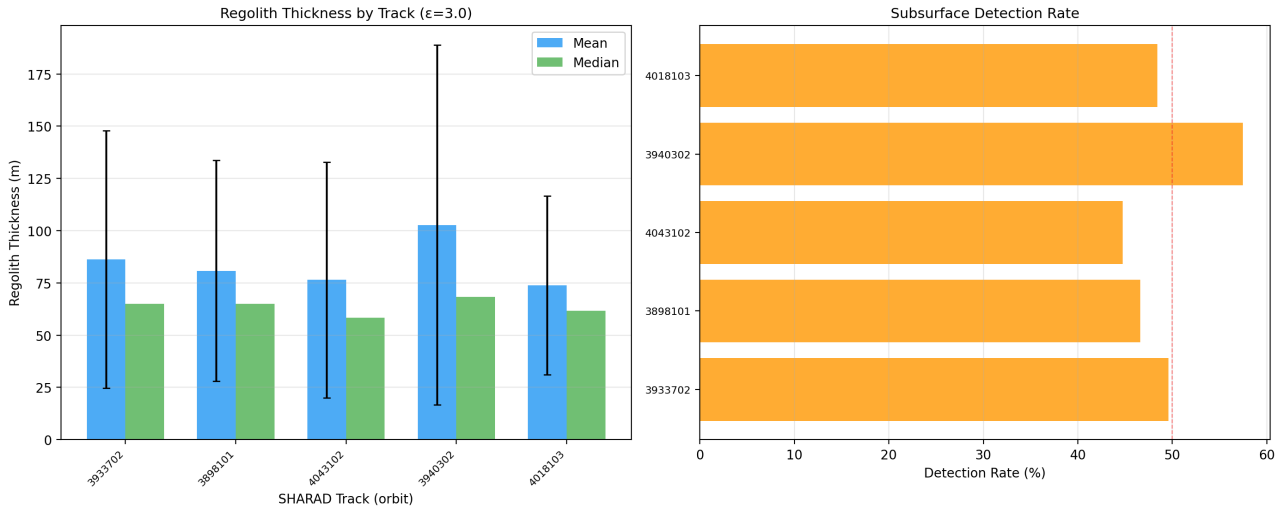


3. Regolith Thickness Estimation (RTE)

Tracks analyzed: 5 Assumed ϵ_r : 3.0 Aggregate mean thickness: 84.0 m Aggregate median thickness: 63.6 m Aggregate detection rate: 49.4%

Track	Mean (m)	Median (m)	Std (m)	Det. Rate	SNR
R_3933702_001_SS19_7	86.3	64.9	61.6	49.6%	18.1
R_3898101_001_SS19_7	80.8	64.9	52.9	46.6%	20.1
R_4043102_001_SS19_7	76.5	58.4	56.4	44.8%	27.9
R_3940302_001_SS19_7	102.7	68.2	86.1	57.5%	13.7
R_4018103_001_SS19_7	73.9	61.7	42.8	48.4%	24.3

Arcadia Planitia — Regolith Thickness Estimation (5 SHARAD tracks)

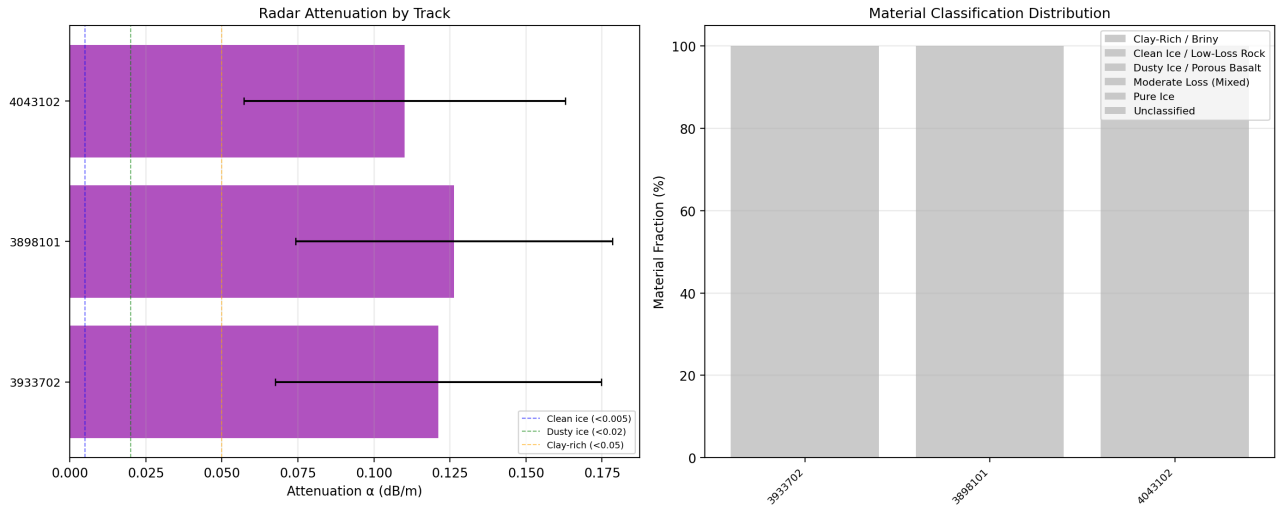


4. Radar Attenuation & Material Classification

Tracks analyzed: 3

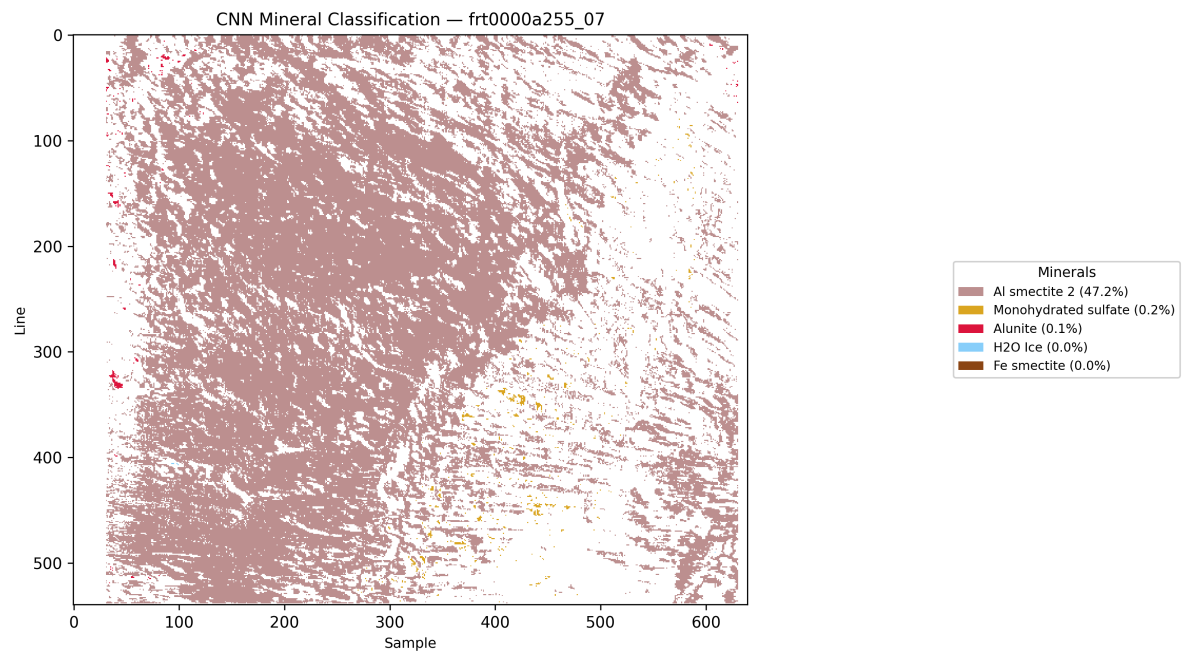
Track	α (dB/m)	Dominant Material	Det. Rate
R_3933702_001_SS19_7	0.1212	Clay-rich / briny	42.0%
R_3898101_001_SS19_7	0.1263	Clay-rich / briny	40.0%
R_4043102_001_SS19_7	0.1101	Clay-rich / briny	37.2%

Arcadia Planitia — Radar Attenuation & Material Classification

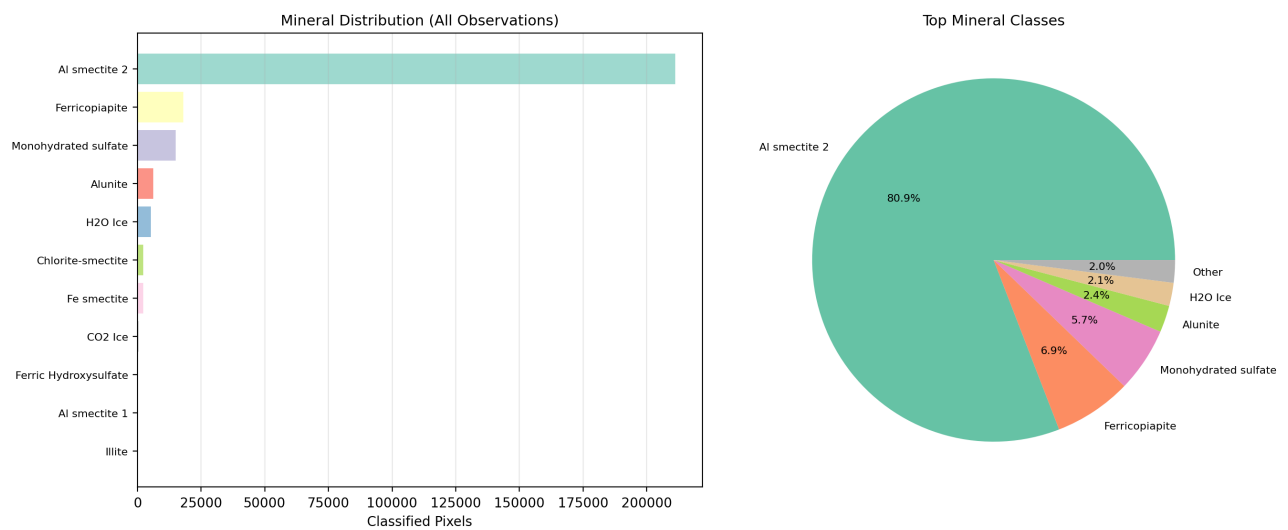


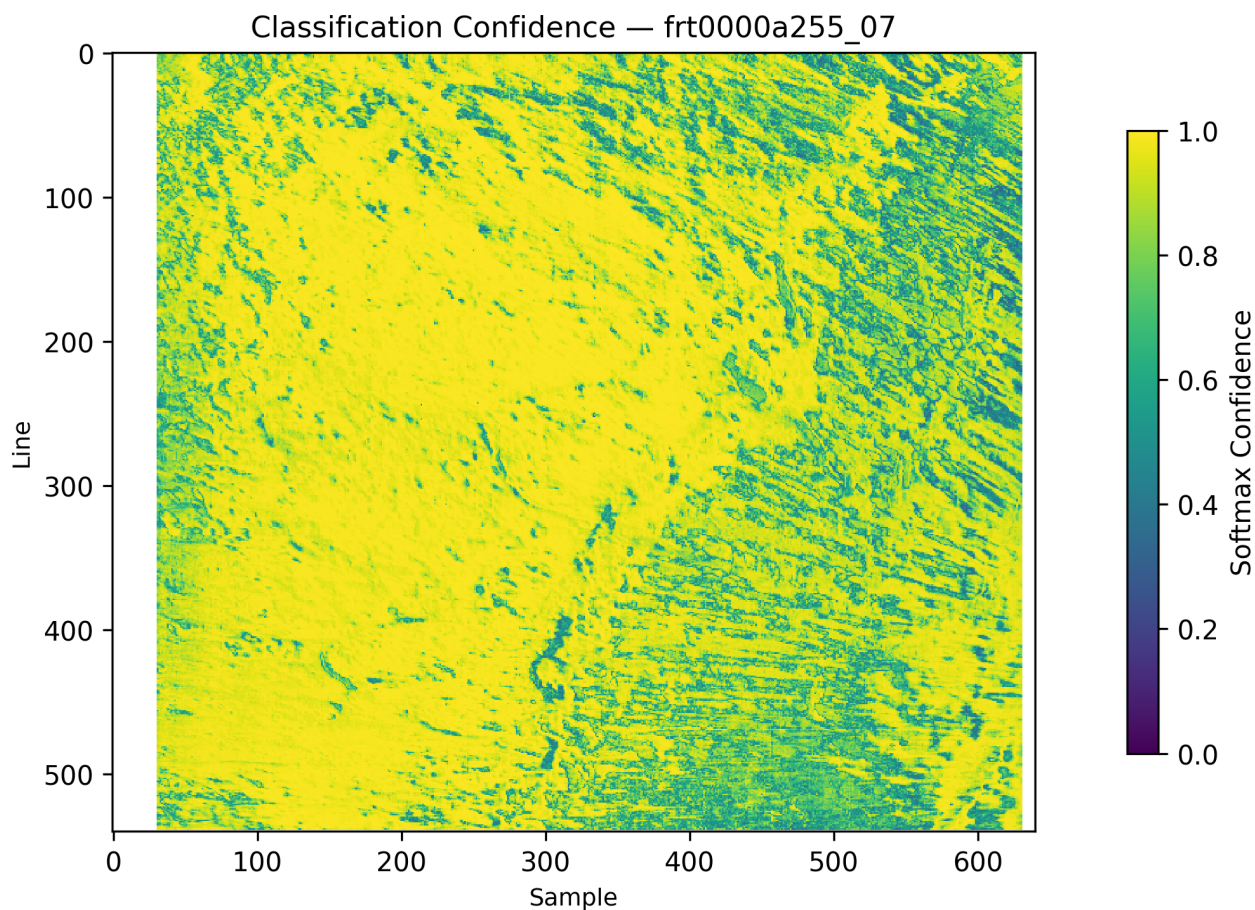
5. CRISM Mineral Classification (CNN)

Observations with CNN results: 4



Arcadia Planitia — CRISM CNN Mineral Classification (4 observations)



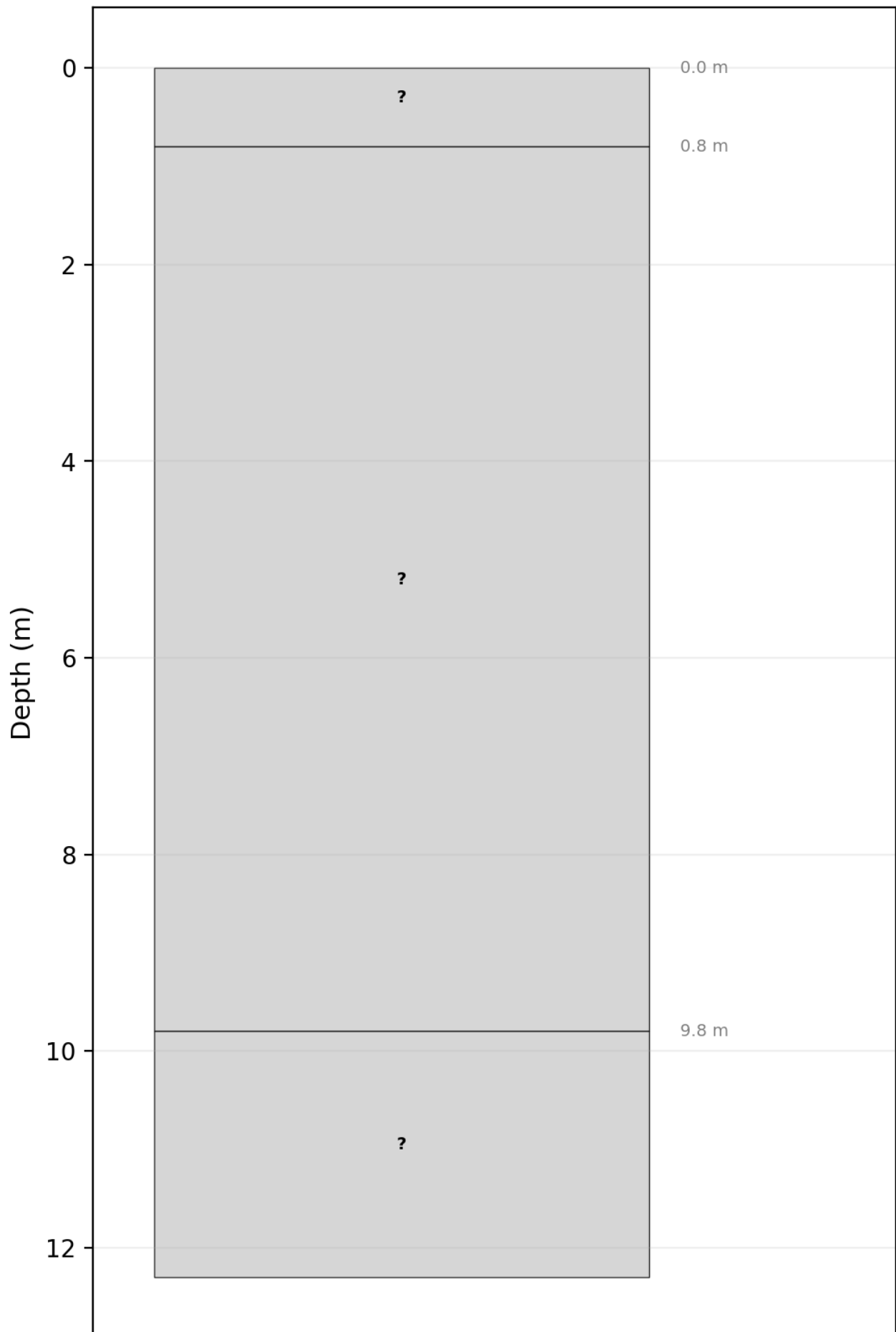


6. Integrated Stratigraphic Column

Location: 47.2°N, -166.3°E (D=20.0 km) **Layers:** 3 **Total depth:** 12.3 m **Instruments:** HiRISE

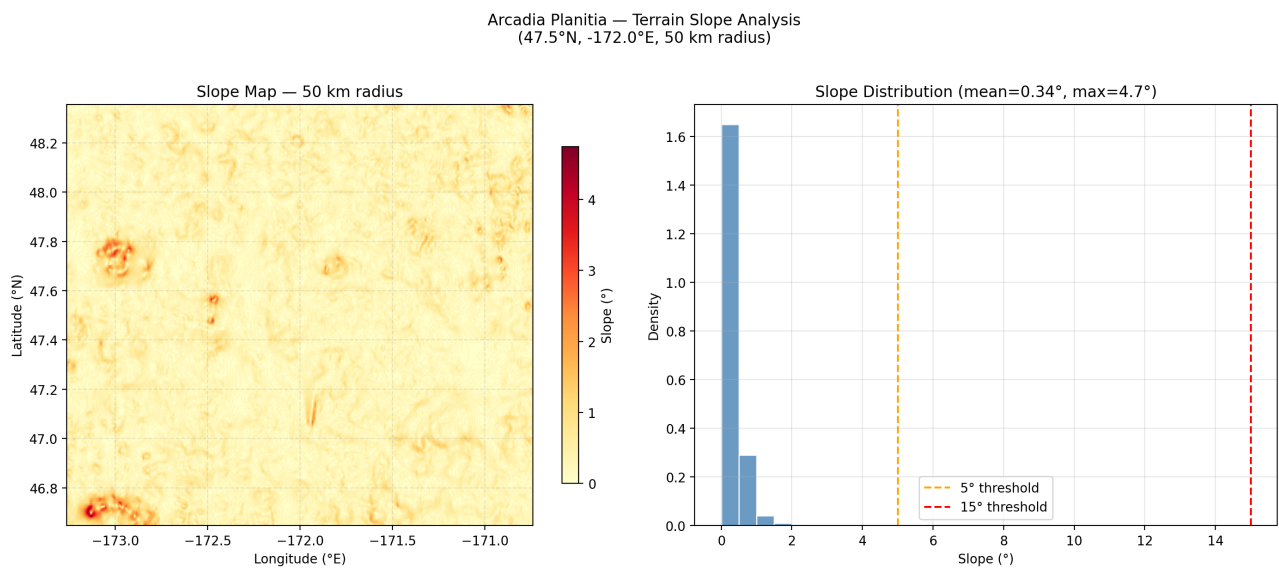
Depth Top (m)	Depth Bottom (m)	Thickness (m)	Mineral	ϵ_r
0.0	0.8	0.8	None	-
0.8	9.8	9.0	None	-
9.8	12.3	2.5	None	-

Stratigraphic Column
(47.2°N, -166.3°E, D=20.0 km)

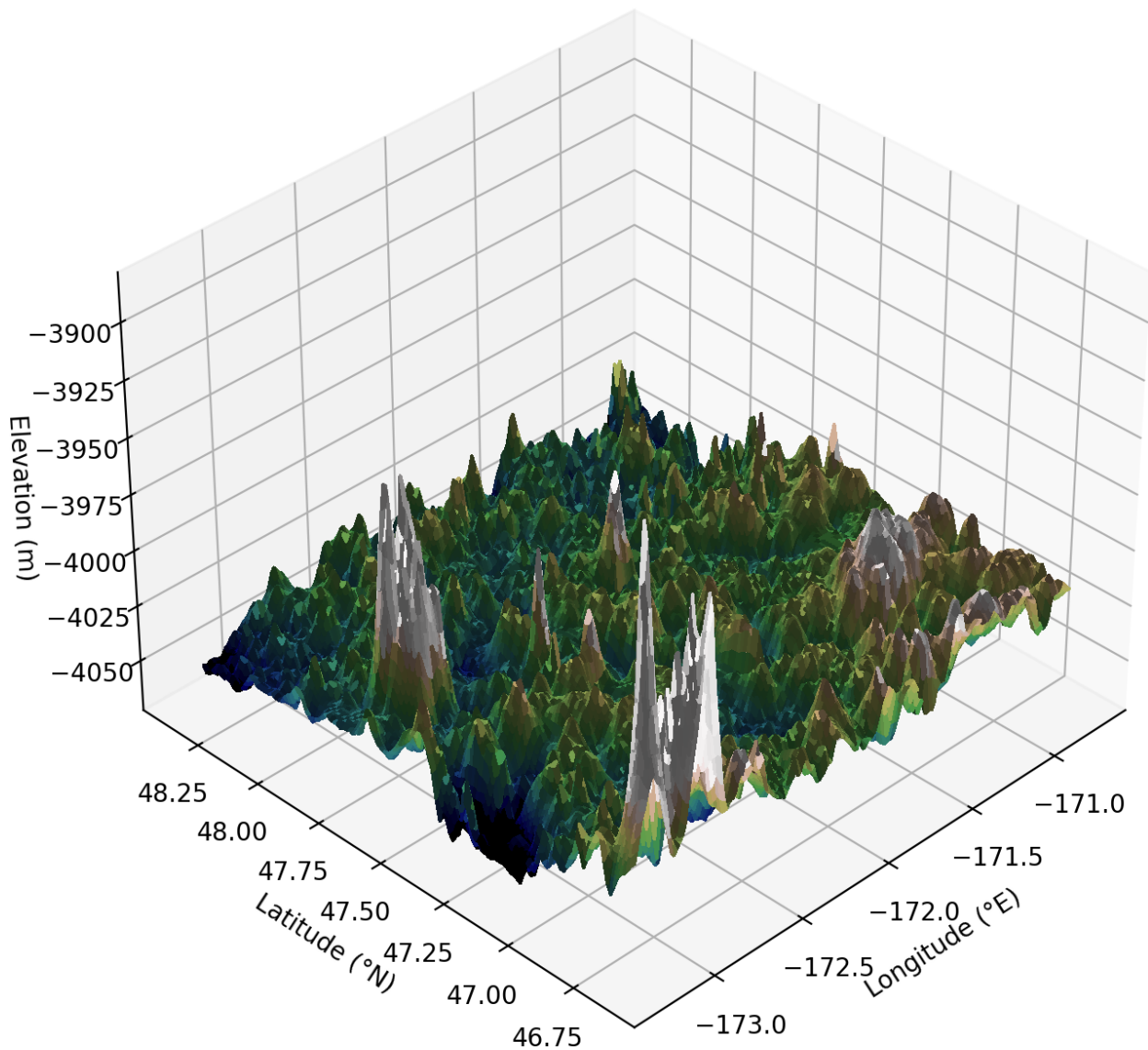


7. Terrain Slope Analysis

Mean slope: 0.34° Max slope: 4.74° Std slope: 0.30° >5° fraction: 0.0% >15° fraction: 0.0%



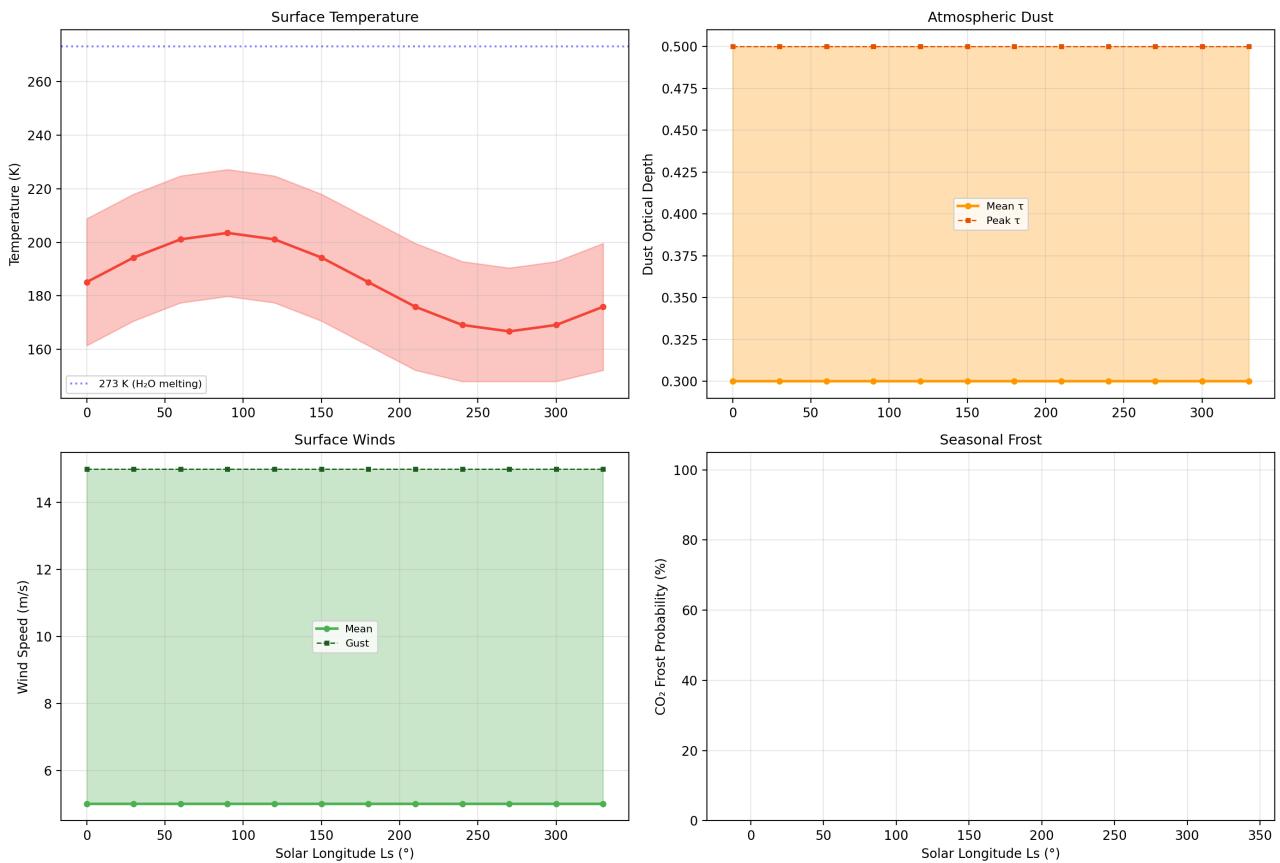
3D Terrain — Arcadia Planitia Relief: 191 m



8. Mars Climate Model

Climate score: 8/10 **Mean temperature:** 203.5 K (-69.6 °C) **Temperature range:** 179.8 — 227.2 K **Surface pressure:** 636 Pa
Dust opacity: τ_{mean} = 0.32, storm risk = LOW **Wind:** mean = 5.4 m/s, gust = 13.4 m/s **CO₂ frost:** probability = 0.0%, seasonal = False

Arcadia Planitia Climate Profile — 47.5°N, -172.0°E Climate Score: 8/10



9. Synthesis & Conclusions

Key Findings

- **Landforms:** 170 features detected across 200 km radius, including 3 craters, 62 channels, 98 ridges, and 0 LDAs.
- **Subsurface:** SHARAD interface detected with 81.3% rate (high confidence). Porous Ice model ($\epsilon=2.80$) gives mean depth ~127 m (range 77-410 m).
- **Regolith:** Mean thickness 84.0 m across 5 tracks ($\epsilon=3.0$), detection rate 49.4%.
- **Material:** Dominant subsurface material classified as **Clay-rich / briny** from radar attenuation analysis.
- **Climate:** Score 8/10. Mean temperature 204 K, CO₂ frost probability 0%.

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